Cascaded $\chi^{(2)}$:$\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in a Doubly-Resonant Dielectric Metasurface

Simulation Report

October 6, 2026

Abstract

A numerical companion to `Cascaded_chi3_metasurface_note.tex`. Twelve figures generated by the modular simulation in `cascaded_chi3/` demonstrate every phenomenon predicted by the temporal coupled-mode theory (TCMT) of Section 5 of the note: the adiabatic effective Kerr; the cascaded two-photon-like loss; sign tunability through $\Delta\omega = 0$; the bandwidth–enhancement sum rule; optical bistability under CW pumping; pump depletion and the parametric-oscillation threshold; the time-domain bandwidth limit; the Manley–Rowe / energy-conservation check; the bulk-PPLN-vs-metasurface trade-off; the geometric overlap as the binding constraint; the $1/n_2^2$ index scaling; and the experimentally observable power-resolved transmission lineshape. We close with concrete design recommendations.

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1 Theoretical setup

The metasurface supports two co-localised resonances:

- FM at ω_1 – quasi-BIC with controlled radiative coupling $\gamma_{1,\mathrm{rad}}$;
- SH at $\omega_2 \approx 2\omega_1$ – symmetry-protected BIC (or under-coupled qBIC), $\gamma_2 \simeq \gamma_2^{\mathrm{nr}}$, large $Q_2 = \omega_2/\gamma_2$.

In the rotating frame at the pump frequency ω_p, with $\delta_p = \omega_p - \omega_1$, $\Delta\omega = 2\omega_1 - \omega_2$ and $\Delta_{\mathrm{SH}} = 2\delta_p + \Delta\omega$,

$$\frac{da_1}{dt} = \big(i\delta_p - \tfrac{\gamma_1}{2}\big)a_1 + i\beta^*\, a_1^*\, a_2 + i\alpha_3|a_1|^2 a_1 + \sqrt{\gamma_{1,\mathrm{rad}}}\, s_+, \tag{1}$$

$$\frac{da_2}{dt} = \big(i\Delta_{\mathrm{SH}} - \tfrac{\gamma_2}{2}\big)a_2 + i\beta\, a_1^2, \tag{2}$$

with input–output relation $s_- = -s_+ + \sqrt{\gamma_{1,\mathrm{rad}}}\, a_1$.

The tensor coupling (note's eq. (beta)) is

$$\beta = \frac{\varepsilon_0\, \omega_1}{N_2} \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}(\mathbf{r})\, f_{1,j}(\mathbf{r})\, f_{1,k}(\mathbf{r})\; d^3r, \tag{3}$$

with N_2 the projection norm of the SH mode ($N_2 = \frac{1}{2}$ for energy-normalised modes, so the prefactor is $2\varepsilon_0\omega_1$). This corrects the $\varepsilon_0\omega_1/4$ of earlier versions, which made β, the bare α_3 and hence $\chi^{(3)}_{\mathrm{casc}}$ and every R 4× too small (note §5). The TCMT simulations are dimensionless ($\beta = 1$) and are unaffected; only the physical-unit conversions (the values of R) change.

Adiabatic elimination. For $|\Delta\omega| \gg \gamma_2/2$ and $|\dot{a}_2| \ll |\Delta\omega\, a_2|$, set $da_2/dt = 0$ in (2) to obtain $a_2 = i\beta\, a_1^2/(\gamma_2/2 - i\Delta_{\mathrm{SH}})$. Substituting into (1) yields an effective Kerr-like nonlinearity with a real (Kerr) and imaginary (loss) part:

$$\alpha_3^{\mathrm{eff,casc}} = -\frac{|\beta|^2 \Delta_{\mathrm{SH}}}{\Delta_{\mathrm{SH}}^2 + (\gamma_2/2)^2}, \qquad \gamma_{2\mathrm{PA}}^{\mathrm{eff}} = \frac{|\beta|^2 (\gamma_2/2)}{\Delta_{\mathrm{SH}}^2 + (\gamma_2/2)^2}. \tag{4}$$

Far off-resonance, $\alpha_3^{\mathrm{eff,casc}} \to -|\beta|^2/\Delta_{\mathrm{SH}}$.

Final boxed cascaded susceptibility. Inverting the definition of α_3 from the FM mode normalisation and using $|\mathbf{f}_n|^2 \sim 1/(2\varepsilon_0 n_n^2\, V_{\mathrm{eff}})$ (note §6):

$$\chi_{\mathrm{casc}}^{(3)} \simeq -\frac{2\,\omega_1}{3\, n_2^2\, \Delta\omega}\, |\chi^{(2)}|^2\, \eta_{\mathrm{geom}}, \tag{5}$$

with $\eta_{\mathrm{geom}} \in [10^{-3}, 0.5]$ a purely geometric overlap. Only n_2 (the SH-mode effective index) survives – the four powers of n_1 cancel between numerator and denominator.

Dimensionless units used in the simulations. $\gamma_2 = 1$ sets the unit of frequency; $\beta = 1$ sets the energy unit. Time is in $1/\gamma_2$, mode amplitudes in $\gamma_2/|\beta|$. The TCMT parameters `Domega`, `delta1`, `gamma1`, `gamma1_rad`, `alpha3` and the input $|s_+|^2$ are then all dimensionless. Where physical units appear (Sections 2.3, 5.1, 5.2, 5.3), we use the $LiNbO_3$ reference of note §10: $\omega_1 = 1.22{\times}10^{15}$ rad/s, $|\chi^{(2)}| = 5{\times}10^{-11}$ m/V, $|\chi^{(3)}|_{\mathrm{native}} = 2{\times}10^{-21}$ m^2/V^2, $n_1 = 2.21$, $n_2 = 2.26$, $\eta_{\mathrm{geom}} = 0.3$.

2 Validation and basic phenomena

2.1 Validation: TCMT versus analytic effective coefficients

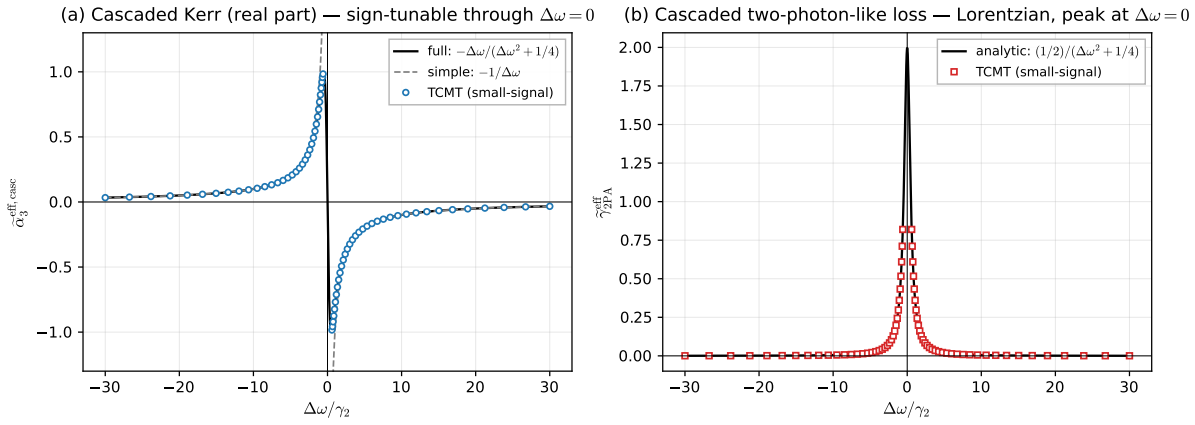


Figure 1. Cascaded effective coefficients extracted from steady-state TCMT (open markers) compared with closed-form analytic curves (black solid). (a) real (Kerr) part; (b) imaginary (two-photon-like loss) part.

Parameters. $\gamma_1/\gamma_2 = 10$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$, $\delta_p = 0$; $|s_+| = 10^{-2}$ (small-signal); sweep $\Delta\omega/\gamma_2 \in [-30, 30]$.

Method. At each detuning, integrate to steady state and extract the cascaded contribution from

$$C \equiv \frac{i\, a_1^* a_2}{a_1\, |a_1|^2} = -\frac{1}{\gamma_2/2 - i\Delta\omega} \quad \Longrightarrow \quad \alpha_{3,\,\mathrm{num}}^{\mathrm{eff,casc}} = \mathrm{Im}(C), \quad \gamma_{2\mathrm{PA},\,\mathrm{num}}^{\mathrm{eff}} = -\mathrm{Re}(C). \tag{6}$$

Panel (a): cascaded Kerr. Numerical TCMT sits exactly on the analytic full form $\alpha_3^{\mathrm{eff,casc}} = -\Delta\omega/(\Delta\omega^2+1/4)$. The simple $-1/\Delta\omega$ form deviates as $|\Delta\omega| \to \gamma_2/2$ – the adiabatic-elimination breakdown of note §6(i) and §13(1).

Panel (b): cascaded TPA. A Lorentzian peaked at $\Delta\omega = 0$, FWHM $= \gamma_2$. The loss is the unavoidable price of resonant enhancement, set entirely by γ_2 with no analogue in $\mathrm{Im}\,\chi^{(3)}$.

Takeaway. TCMT matches the closed-form expressions to four decimals. The simple $-1/\Delta\omega$ form, common in cascading literature, underestimates the true Kerr coefficient by 70 % at $\Delta\omega = \gamma_2/2$ and the gap grows fast as resonance is approached.

2.2 Sign tunability – walking $\Delta\omega$ through zero

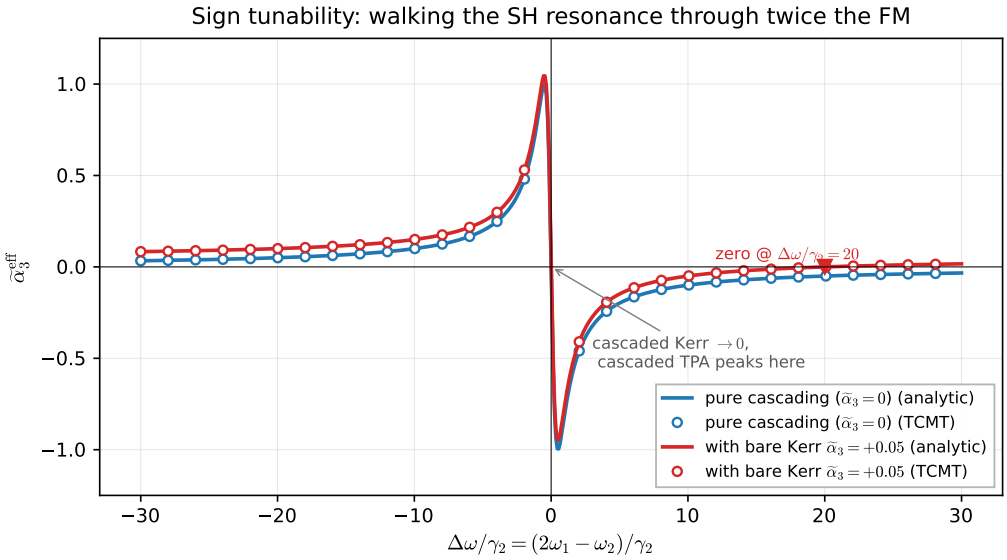


Figure 2. Effective Kerr coefficient versus SH detuning. Blue: pure cascading. Red: cascading plus a positive bare bulk Kerr; the zero of the total response shifts to $\Delta\omega/\gamma_2 = 1/\tilde{\alpha}_3 = 20$. Grey shading marks the adiabatic-elimination breakdown band.

Parameters. $\gamma_1/\gamma_2 = 10$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\delta_p = 0$; two scenarios for the bare Kerr: $\tilde{\alpha}_3 \in \{0, +0.05\}$; sweep $\Delta\omega/\gamma_2 \in [-30, 30]$ (excluding $|\Delta\omega| < \gamma_2$).

Why it matters. The cascaded Kerr is odd in $\Delta\omega$, $\alpha_3^{\mathrm{eff,casc}}(-\Delta\omega) = -\alpha_3^{\mathrm{eff,casc}}(\Delta\omega)$. Walking the SH resonance through $2\omega_1$ – by tuning the meta-atom geometry, applying strain, or thermal tuning – flips the sign of the effective Kerr from self-defocusing ($\Delta\omega > 0$) to self-focusing ($\Delta\omega < 0$). With a non-zero bare Kerr, the total zero shifts to $\Delta\omega = 1/\tilde{\alpha}_3$.

Takeaway. A single metasurface can deliver self-focusing, self-defocusing, or cancel native Kerr exactly, by choice of $\Delta\omega$. This was historically exploited in soliton compression experiments (DeSalvo 1992, Liu 1999).

2.3 Bandwidth–enhancement sum rule

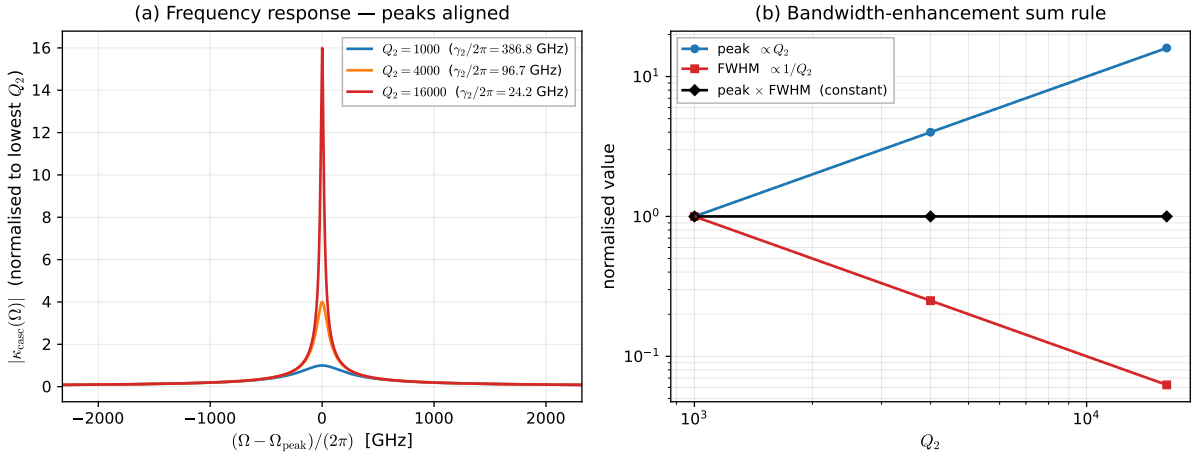


Figure 3. (a) Modulation transfer function $|\kappa_{\mathrm{casc}}(\Omega)|$ evaluated on a shared Ω grid for three Q_2 values, peaks aligned at $\Omega = -\Delta_{\mathrm{SH}}$. (b) Peak grows $\propto Q_2$; FWHM shrinks $\propto 1/Q_2$; their product is Q_2-independent.

Parameters. $\omega_2 = 2\omega_1$, $\omega_1 = 1.22 \times 10^{15}$ rad/s ($LiNbO_3$); $Q_2 \in \{1\,000, 4\,000, 16\,000\}$; $\Delta\omega = \gamma_2(Q_2{=}4\,000)/2$ chosen so the mid curve sits at the bandwidth-optimal point; Ω grid spans $\pm 6\gamma_2^{\max}$ about $\Omega_{\mathrm{peak}} = -\Delta\omega$.

Derivation. The cascaded modulation transfer function is $\kappa_{\mathrm{casc}}(\Omega) = -|\beta|^2/[\gamma_2/2 - i(\Omega + \Delta_{\mathrm{SH}})]$. At the bandwidth-optimal point $\Delta_{\mathrm{SH}} = \gamma_2/2$,

$$\mathrm{peak}\,|\kappa| = \frac{|\beta|^2}{\gamma_2/2} \propto Q_2, \quad \mathrm{FWHM} = \sqrt{3}\,\gamma_2 \propto 1/Q_2, \quad \mathrm{peak} \times \mathrm{FWHM} = 2\sqrt{3}\,|\beta|^2. \tag{7}$$

Numerical verification. Peak ratios [1, 4, 16], FWHM ratios [1, 1/4, 1/16], product ratios [1, 1, 1] to better than 2×10^{-4} across the three Q_2 values.

Takeaway. Resonant enhancement *redistributes* nonlinear action; it does not increase the area under the curve. This is the same rule that limits oscillator-strength sums in linear absorption – purely classical, unbreakable by clever resonator design.

3 Cascaded-induced FM dynamics

3.1 Bistability: the first experimental fingerprint at the FM

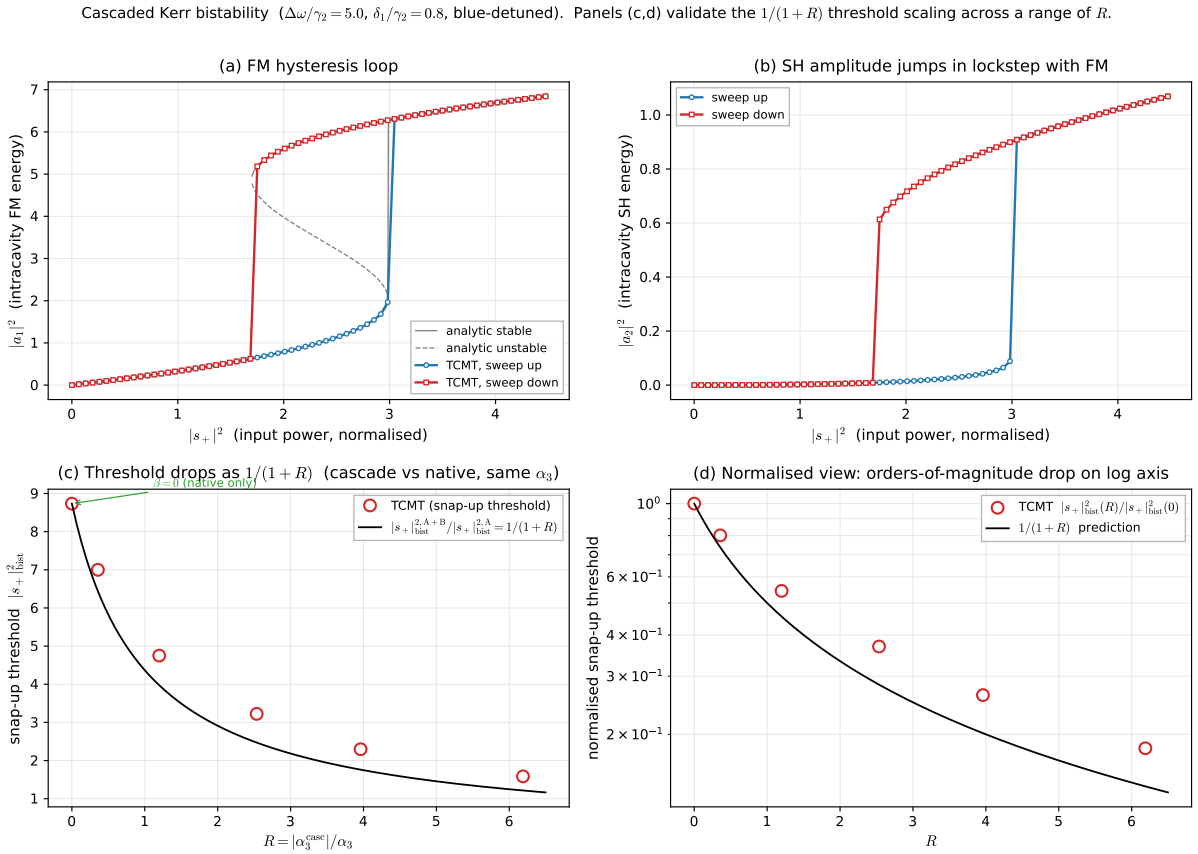


Figure 4. (a) Hysteresis in the intracavity FM energy under a stepwise quasi-static input sweep. The pump is detuned blue of the cold-cavity resonance ($\delta_p > 0$) so that the intensity-induced red-shift of the dressed FM resonance (negative cascaded Kerr at $\Delta\omega > 0$) climbs toward the pump as input grows – the standard geometry for self-defocusing-Kerr bistability. Snap-up at $|s_+|^2 \approx 3.0$, snap-down at ≈ 1.7; both sit on the analytic Kerr+2PA saddle-nodes. (b) SH amplitude jumps in lockstep with the FM, tied by $a_2 = i\beta\, a_1^2/(\gamma_2/2 - i\Delta_{\text{SH}})$. (c,d) Threshold-vs-$R$ sweep at fixed $\Delta\omega$, δ_p, and α_3 (constructive sign): native Kerr ($\beta = 0$) anchors $R = 0$; increasing β adds an aligned cascade contribution α_3^{casc}, raising R. TCMT measurements (red) follow the predicted $1/(1+R)$ scaling (black) to within $\sim$30%; the residual is the cascaded TPA-driven cusp distortion that the pure-Kerr $1/(1+R)$ formula does not capture.

Parameters. $\Delta\omega/\gamma_2 = 5$, $\gamma_1/\gamma_2 = 0.4$, $\gamma_{1,\text{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$, $\delta_p/\gamma_2 = 0.8$ (blue-detuned by $4 \cdot \gamma_1/2$); 70-step quasi-static sweep with state continuation, $|s_+|^2 \in [10^{-3}, 4.5]$, settle time $t_s = 250/\gamma_2$ per step.

Why blue detuning? With $\Delta\omega > 0$, $\alpha_3^{\text{eff,casc}} < 0$: the effective FM resonance shifts down with intensity. To get hysteresis, the pump must sit above the cold-cavity resonance ($\delta_p > 0$); as input grows, the shifted resonance climbs toward the pump, the response runs away, and the system snaps to the upper branch.

Cusp condition. From the steady-state cubic

$$|a_1|^2\left[(g_1 + \gamma_e|a_1|^2)^2 + (\delta_p + \alpha_e|a_1|^2)^2\right] = \gamma_{1,\text{rad}}\,|s_+|^2, \tag{8}$$

with $g_1 = \gamma_1/2$, $\alpha_e = \alpha_3 + \alpha_3^{\text{eff,casc}}$, $\gamma_e = \gamma_{2\text{PA}}^{\text{eff}}$, the saddle-node fold opens for $\delta_p > \sqrt{3}\,g_1$. Here $\delta_p = 4g_1 \gg \sqrt{3}g_1$, comfortably bistable.

Takeaway. A purely monochromatic CW pump can drive the FM through bistability via cascaded SPM. This is the cleanest signature of cascaded $\chi^{(3)}$ at the FM port – far more sensitive than reflectance measurements at the SH wavelength.

3.2 Saturation and pump depletion

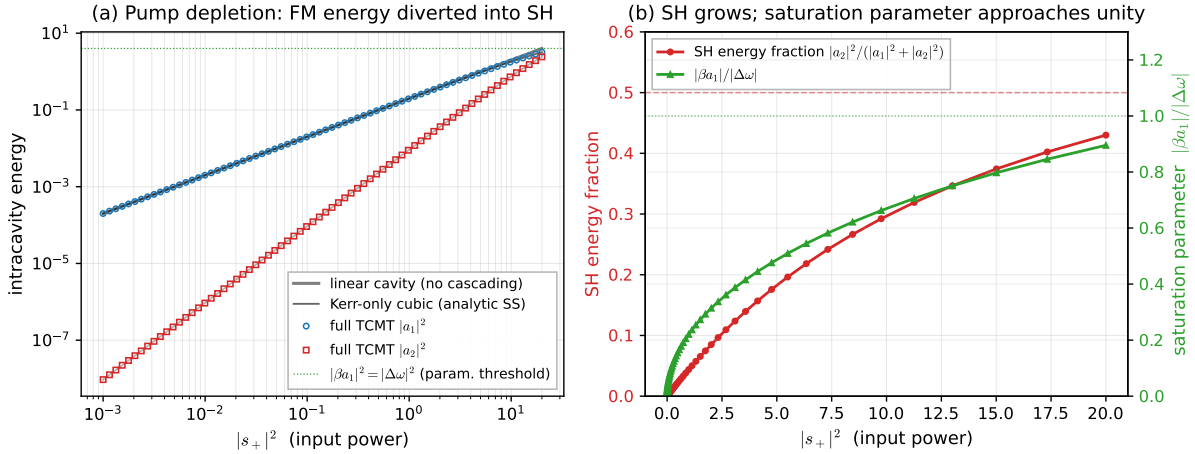


Figure 5. (a) Log–log of intracavity energies. Linear-cavity reference (grey), Kerr-only cubic (black), full TCMT with cascading (open markers). The cubic and the TCMT trivial fixed point are equivalent to machine precision. (b) Linear-x view: SH energy fraction (red, left) and saturation parameter $|\beta a_1|/|\Delta\omega|$ (green, right) climb together.

Parameters. $LiNbO_3$ design point of note §10: $\Delta\omega/\gamma_2 = 2$, $\gamma_1/\gamma_2 = 10$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$, $\delta_p = 0$; 70-step state-continuation sweep $|s_+|^2 \in [10^{-3}, 20]$; settle time $1500/\gamma_2$ per step.

Why the cap at sat ≈ 1*?* Past $|\beta a_1| \gtrsim |\Delta\omega|$ the trivial fixed point becomes parametrically unstable and the deterministic ODE enters a limit cycle (Section 3.3). Below threshold, the Kerr-only cubic and the full TCMT agree to machine precision – the algebraic substitution $a_2 = i\beta\, a_1^2/(\gamma_2/2 - i\Delta_{\mathrm{SH}})$ is exact at any deterministic steady state, with no approximation.

Takeaway. The Kerr-equivalent description holds up to $|\beta a_1| \approx |\Delta\omega|$. Beyond, the FM sees a non-perturbative back-action – useful for parametric oscillation, bad for "the system as a clean Kerr medium". The maximum useful $|a_1|^2$ scales as $1/Q_2$ at fixed $\Delta\omega/\gamma_2$: high-Q devices saturate at *lower* powers.

3.3 Parametric-oscillation onset above threshold

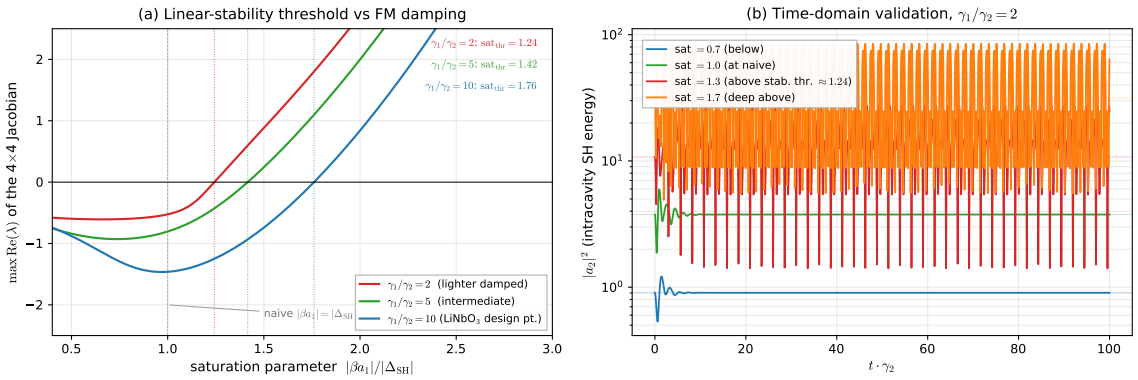


Figure 6. (a) Largest real part of the 4×4 Jacobian eigenvalue of the trivial fixed point versus the saturation parameter, for three FM-damping levels. The crossing of zero is the parametric-oscillation threshold: $\mathrm{sat}_{\mathrm{thr}} = 1.24, 1.42, 1.76$ for $\gamma_1/\gamma_2 = 2$, 5, 10 respectively. The realistic $LiNbO_3$ design point ($\gamma_1/\gamma_2 = 10$) sits at $\mathrm{sat}_{\mathrm{thr}} \approx 1.76$, well above the naive $|\beta a_1| = |\Delta\omega|$ mark. (b) Time-domain validation at the lighter-damped $\gamma_1/\gamma_2 = 2$, chosen so the threshold sits at numerically tractable input. Below threshold, a 10^{-3} SH perturbation decays back to the trivial FP; above threshold, the trajectory enters a clear limit cycle. Dotted lines show the trivial-FP values of matching colour.

Parameters. $\Delta\omega/\gamma_2 = 2$, $\gamma_1/\gamma_2 = 2$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$, $\delta_p = 0$. Lighter FM damping than the realistic design point so the threshold sits at numerically tractable input. Four sat parameters $\{0.7, 1.0, 1.3, 1.7\}$, integrated for $100/\gamma_2$ from $a_2(0) = a_2^{\mathrm{triv}} + 10^{-3}\, i$.

Interpretation. Above threshold, the trivial fixed point $a_2 = i\beta a_1^2/(\gamma_2/2 - i\Delta_{\mathrm{SH}})$ has at least one positive linear-stability eigenvalue. Any perturbation with non-zero projection on the unstable mode grows exponentially until the nonlinearity clamps it onto a limit cycle. This is the analogue of the OPO threshold of note §13(2). The deterministic "fixed point" the saturation figure plots is metastable.

Takeaway. $|\beta a_1| \approx |\Delta\omega|$ is a hard ceiling for the cascaded-Kerr description – not because the cubic equation diverges (it does not), but because the deterministic fixed point loses stability and the system runs off to a parametric-oscillator regime no Kerr medium can model.

4 Time-domain consequences

4.1 Pulse-bandwidth limit

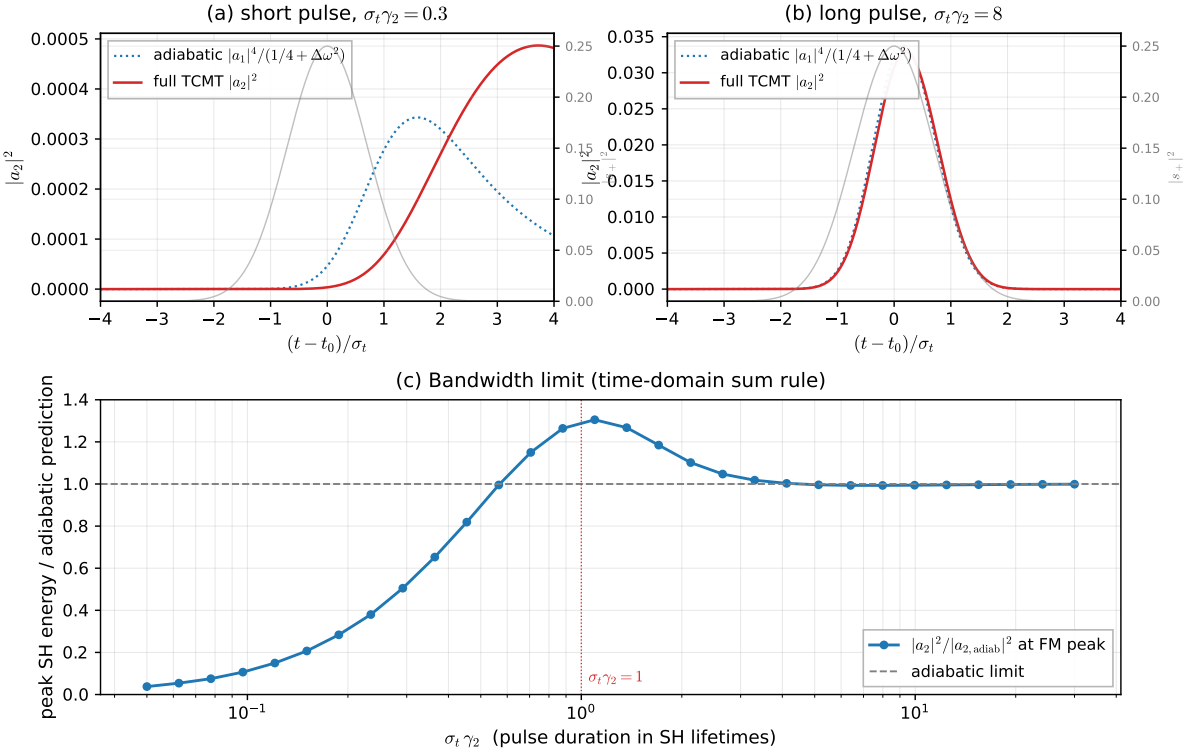


Figure 7. (a) Short pulse $\sigma_t\gamma_2 = 0.3$: the SH lags and undershoots the adiabatic prediction (blue dotted), then rings down at rate $\gamma_2/2$. (b) Long pulse $\sigma_t\gamma_2 = 8$: the SH faithfully tracks $|a_1|^2$. (c) Tracking quality at the moment of FM peak across $\sigma_t\gamma_2 \in [0.05, 30]$. The ratio crosses unity at $\sigma_t\gamma_2 \approx 1$ – the time-domain face of the bandwidth–enhancement sum rule – and overshoots to ~ 1.3 near $\sigma_t\gamma_2 = 1$ before settling. The overshoot is not numerical ringing: at marginal pulse duration the FM peaks just before the SH catches up, and the SH receives a brief impulsive drive that pushes its peak above the static adiabatic value $|a_1|^4/(\Delta\omega^2 + (\gamma_2/2)^2)$ for an instant.

Parameters. $\Delta\omega/\gamma_2 = 2$, $\gamma_1/\gamma_2 = 1$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$, $\delta_p = 0$; Gaussian pump $s_+(t) = s_0 \exp[-(t-t_0)^2/(2\sigma_t^2)]$ with $s_0 = 0.5$ for panels (a)–(b), $s_0 = 0.3$ for the duration sweep in panel (c).

Adiabatic prediction. Whenever the SH stays slaved to the FM, $|a_2(t)|^2 = |a_1(t)|^4/[\Delta\omega^2 + (\gamma_2/2)^2]$.

Takeaway. Cascaded $\chi^{(3)}$ enhancement is not "fast" Kerr in the ultrafast sense. For pulse durations shorter than $\sim 1/\gamma_2$ the SH cannot follow and the cascaded contribution rolls off. With $Q_2 = 4\,000$, $1/\gamma_2 \approx 1.7$ ps; anything sub-picosecond is too fast.

4.2 Manley–Rowe / energy-conservation check

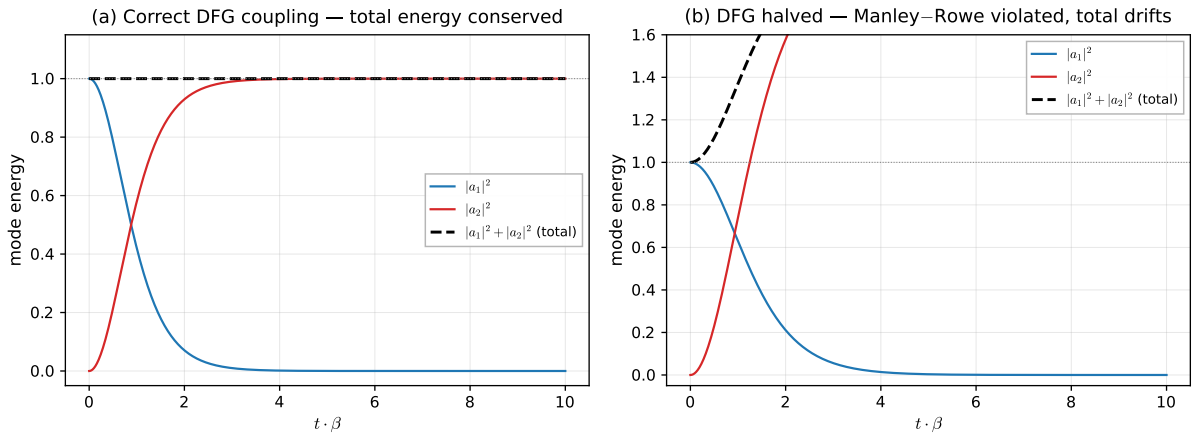


Figure 8. Lossless TCMT, no input. (a) Correct DFG factor: total energy stays at 1.0 to machine precision. (b) Halved DFG (the "κ vs $\kappa/2$" bug of the original derivation): total energy grows past 1.6 by $t = 2/\beta$ – energy created out of nothing.

Parameters. $\gamma_1 = \gamma_2 = 0$ (lossless), $s_+ = 0$ (no input), $\Delta\omega = 0$ (degenerate, fastest exchange); initial state $(a_1, a_2) = (1, 0)$; integration window $t \cdot \beta \in [0, 10]$.

Why this matters. Manley–Rowe in the lossless limit demands $\frac{d}{dt}(|a_1|^2 + |a_2|^2) = 0$, equivalently $\beta^{(1)} = \beta^*$: a single coupling constant on both sides, not two independent ones. This is built in by the factor of 2 in the DFG polarisation $P^{(2)}(\omega_1) = 2\varepsilon_0\,\chi^{(2)}\,E(2\omega_1)\,E^*(\omega_1)$ that note §2 rederives.

Numerical drifts. Correct: total-energy drift over $10/\beta$ time units is 1.4×10^{-14} (machine precision). Halved DFG: drift $= 1.0$, the same order of magnitude as the total energy itself.

Takeaway. The factor-of-2 the note rederives is not a cosmetic fix – without it, the model violates the second law. Any TCMT code that drops it is silently wrong; this figure is the smoke test.

5 Design space

5.1 Bulk PPLN versus metasurface: the same sum rule, different operating point

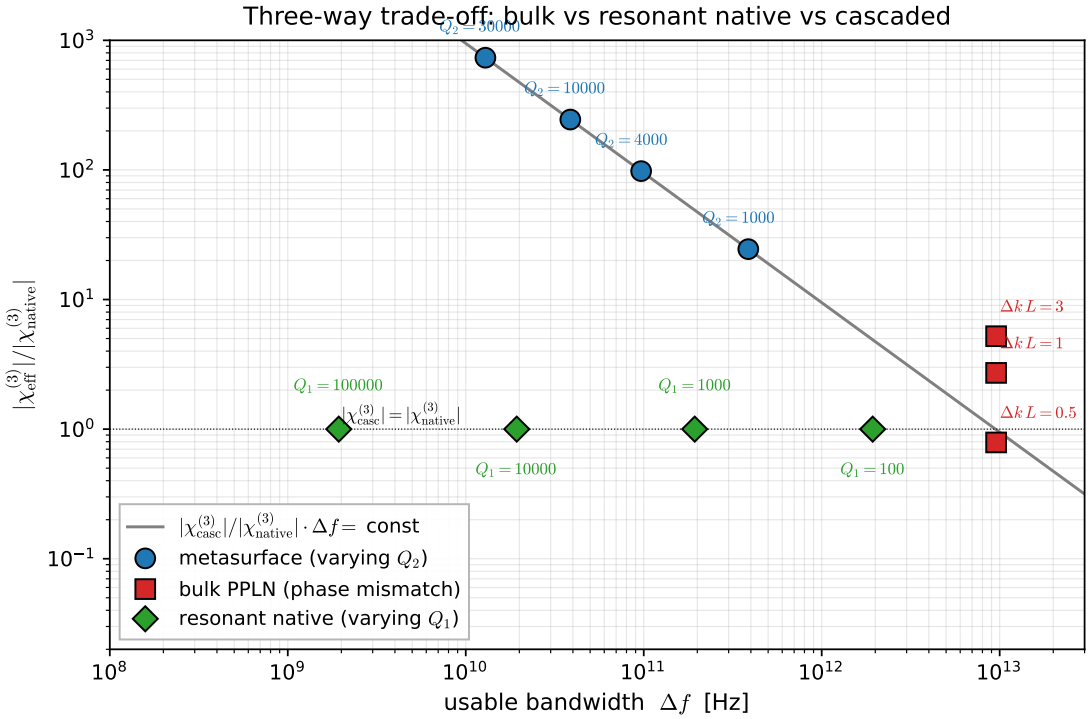


Figure 9. Enhancement $|\chi^{(3)}_{\rm casc}|/|\chi^{(3)}_{\rm native}|$ versus usable bandwidth in lab units. Circles: doubly-resonant metasurface, varying Q_2. Squares: bulk phase-mismatched PPLN. Both regimes lie on the common sum-rule hyperbola defined by the same $|\chi^{(2)}|^2\eta_{\rm geom}/n_2^2$.

Parameters. $LiNbO_3$: $\omega_1 = 1.22\times10^{15}$ rad/s, $|\chi^{(2)}| = 5\times10^{-11}$ m/V, $|\chi^{(3)}|_{\rm native} = 2\times10^{-21}$ $\mathrm{m}^2/\mathrm{V}^2$, $n_2 = 2.26$, $\eta_{\rm geom} = 0.3$. Metasurface points: $Q_2 \in \{10^3, 4\times10^3, 10^4, 3\times10^4, 10^5\}$. Bulk PPLN points: $\Delta k\,L \in \{0.5, 1, 3\}$, $\Delta f \approx 1$ THz.

Formulas. The metasurface enhancement, evaluated at fixed $\Delta\omega/\gamma_2$ via eq. (5):

$$\mathrm{enh}(\Delta\omega/\gamma_2) = \frac{Q_2}{n_2^2}\,\frac{(\Delta\omega/\gamma_2)^{-1}}{3/2}\,\frac{\eta_{\rm geom}\,|\chi^{(2)}|^2}{|\chi^{(3)}|_{\rm native}}, \quad \Delta f = \frac{\gamma_2}{2\pi} = \frac{2\omega_1}{2\pi Q_2}. \tag{9}$$

Two natural operating points (Table 1): the bandwidth-optimal $\Delta\omega = \gamma_2/2$ (peak of $|\alpha_3^{\rm eff,casc}|$ in Fig. 1a) and the "conservative" $\Delta\omega = 2\gamma_2$, where saturation is pushed up by $4\times$ at the cost of $4\times$ lower enhancement. The bulk regime uses the SVEA Stegeman/DeSalvo result

$$\chi^{(3)}_{\rm casc,bulk} = -\frac{2\,\omega_1}{3\,c\,n_2\,\Delta k}\,|\chi^{(2)}|^2\,F(\Delta k\,L), \quad F(x) \equiv 1 - \frac{\sin x}{x}\,e^{-ix}, \tag{10}$$

with bandwidth $\Delta f \sim |\Delta k|\,v_g/(2\pi) \sim$ few THz. (Note: only the SH-mode index n_2 appears; the powers of n_1 from the FM SVEA projection cancel against n_1 in the cascade kernel, paralleling the metasurface single-resonance result. See main note eq. (chi3casc-bulk).)

$\Delta\omega/\gamma_2$	prefactor	$R(Q_2{=}4\,000)$	$R(Q_2{=}10^4)$	Δf [GHz]
1/2 (BW-optimal)	$Q_2/(3\,n_2^2)$	98	245	97
1 (edge of adiabatic)	$0.8\,Q_2/(3\,n_2^2)$	78	196	97
2 (conservative)	$0.47\,Q_2/(3\,n_2^2)$	46	115	97

Table 1. Cascaded-$\chi^{(3)}$ enhancement at three operating points, $LiNbO_3$ at telecom ($n_2 = 2.26$, $\eta_{\rm geom} = 0.3$, $|\chi^{(2)}| = 5{\times}10^{-11}$ m/V, $|\chi^{(3)}|_{\rm native} = 2{\times}10^{-21}$ m^2/V^2, $\gamma_2 = \omega_2/Q_2 = 2\omega_1/Q_2$). Prefactors and R use the full Lorentzian $\Delta\omega/[\Delta\omega^2 + (\gamma_2/2)^2]$; the far-detuned form $1/\Delta\omega$ used in the note's worked example gives $Q_2/(6n_2^2)$ and $R \simeq 49$ at $\Delta\omega = 2\gamma_2$ (within 6%) but would double the $\Delta\omega = \gamma_2/2$ value, where it is not valid. (Earlier versions of this table used the far-detuned form throughout and carried the factor-4 prefactor error, quoting 49/25/12.) $\Delta f = \gamma_2/(2\pi)$ quoted at $Q_2 = 4\,000$; it depends on Q_2, not on $\Delta\omega$. The summary table and Fig. 9 use the conservative point throughout for consistency.

Sum-rule constant from the metasurface family at the conservative operating point: enh$\cdot\Delta f$(Hz) $\simeq$ $4.7{\times}10^{12}$. Pulling $\Delta\omega$ down to γ_2 doubles the constant (at the cost of edge-of-adiabatic operation); pulling further to $\gamma_2/2$ doubles again. The hyperbola is the same; only its absolute height shifts.

Trade-off. With the corrected $1/n_2$ scaling, the bulk PPLN $|\chi^{(3)}_{\rm casc}|$ in $LiNbO_3$ at $\Delta k{=}3{\times}10^5$ m^{-1} is $\sim 10^{-20}$ m^2/V^2, roughly 5$\times$ native – not "comparable" to native as the earlier draft claimed. The metasurface at $Q_2 = 4{,}000$ delivers $\sim 100\times$ native at the bandwidth-optimal point ($\Delta\omega = \gamma_2/2$) or $\sim 49\times$ at the conservative operating point ($\Delta\omega = 2\gamma_2$), so the metasurface wins the $|\chi^{(3)}|$ comparison by $\sim$ 10–20$\times$ at routine Q_2. In the $|\chi^{(3)}|\cdot\Delta f$ product, however, bulk leads by $\sim$ 5–10$\times$ because its $\sim$ 10 THz bandwidth dwarfs the cascade's $\lesssim$ 100 GHz bandwidth. The metasurface gains $\sim$ 1 order of magnitude in $|\chi^{(3)}_{\rm eff}|$ and gives up $\sim$ 2 orders of magnitude in bandwidth. Both regimes are best-in-class for their respective applications: high-Q metasurfaces for CW-pumped bistability, parametric thresholds, and frequency-comb stabilisation; bulk phase-mismatched PPLN for ultrafast soliton compression and broadband all-optical switching.

5.2 Geometric overlap is the binding constraint

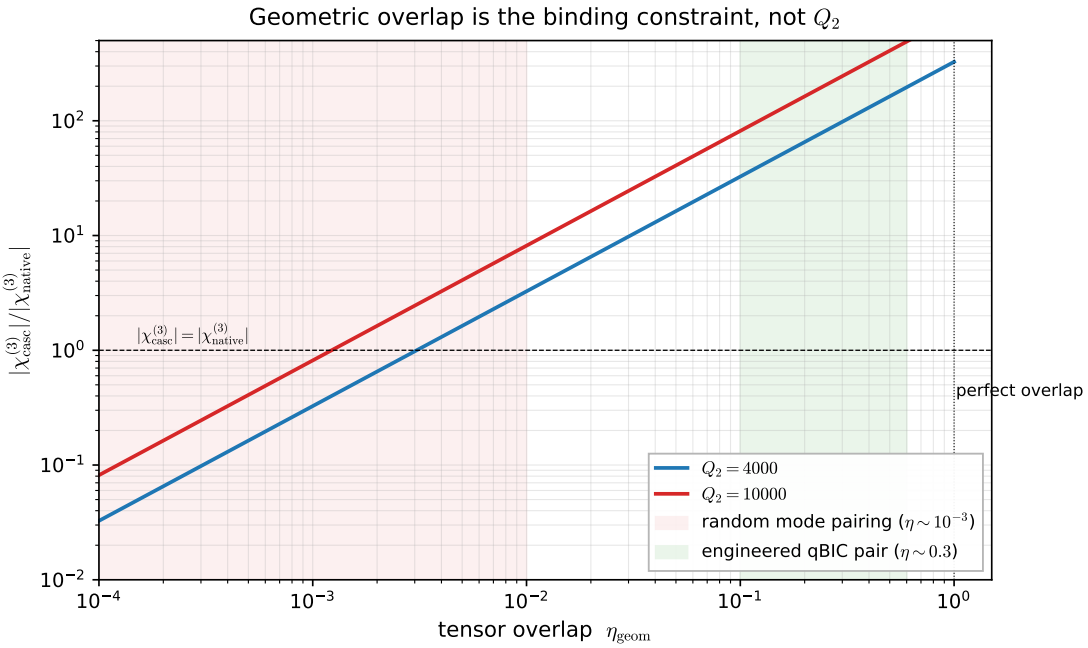


Figure 10. Cascaded enhancement versus tensor overlap $\eta_{\rm geom}$ on log–log, for two Q_2 values. Random mode pairing ($\eta_{\rm geom} \sim 10^{-3}$, red shaded) gives sub-native cascading even at $Q_2 = 10^4$. Engineered qBIC pair ($\eta_{\rm geom} \sim 0.3$, green shaded) recovers $\sim$ 100–250$\times$ enhancement (at $\Delta\omega = \gamma_2$, far-detuned form).

Parameters. $LiNbO_3$ numbers as above; $Q_2 \in \{4\,000,\ 10\,000\}$; $\eta_{\rm geom} \in [10^{-4},\ 1]$ on a log grid; vertical bands at $\eta_{\rm geom} \sim 10^{-3}$ (random pairing) and $\eta_{\rm geom} \sim 0.3$ (engineered qBIC).

Conditions for non-vanishing overlap (note §8). The integral $\eta_{\rm geom} \propto \left| \int \chi^{(2)}_{ijk}(\mathbf{r})\, f^*_{2,i}\, f_{1,j}\, f_{1,k}\, d^3r \right|^2$ vanishes unless three conditions hold simultaneously:

1. No bulk inversion symmetry in the meta-atom material ($LiNbO_3$, GaAs, GaP, AlGaAs all qualify).
2. Mode-symmetry mismatch is lifted: if both modes share parity under all spatial symmetries, the integrand has definite parity that may force the integral to zero. Quasi-BIC engineering breaks one such symmetry while preserving high Q_2.
3. Adequate radial overlap of f_2 with f_1^2 at the unit-cell scale.

Takeaway. Q_2 alone buys nothing if $\eta_{\rm geom}$ is small. A factor-of-2.5 in Q_2 shifts the line by ~ 0.4 dex; a factor-of-100 reduction in $\eta_{\rm geom}$ shifts it 2 dex *down*. Geometry first; resonance Q second.

5.3 Refractive-index dependence: $1/n_2^2$, not $1/n^3$

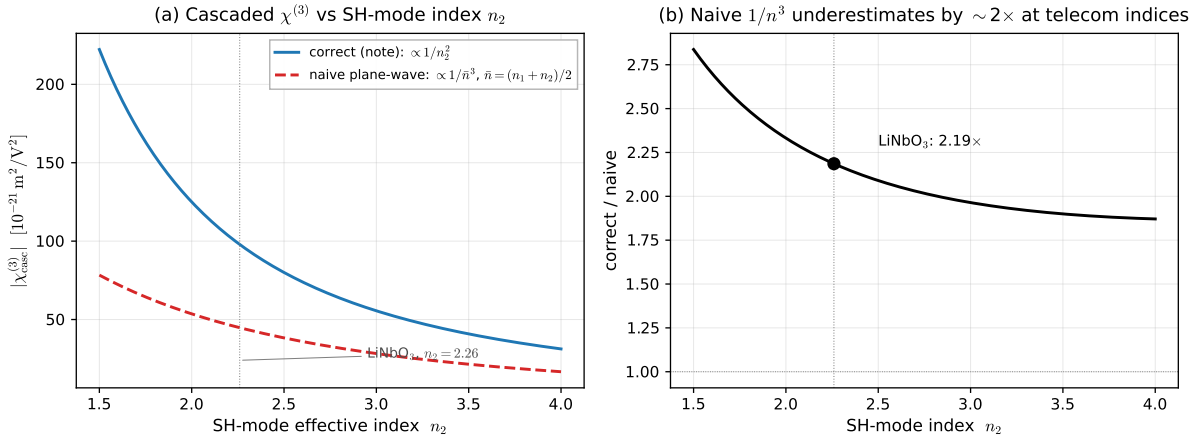


Figure 11. (a) $|\chi^{(3)}_{\rm casc}|$ versus SH-mode index n_2 at fixed $Q_2 = 4\,000$, $\eta_{\rm geom} = 0.3$. The resonant single-mode result (5) (blue) versus the bulk plane-wave $1/\bar{n}^3$ form (red, with $\bar{n} = (n_1 + n_2)/2$). (b) Their ratio. At $LiNbO_3$ ($n_2 = 2.26$) the ratio is 2.19, matching $\bar{n}^3/n_2^2 = 2.18$. The figure is comparing two derivations side-by-side, not validating one against the other: the bulk form is incorrect for resonant single-mode cascading because the four powers of n_1 do not appear in the bulk derivation.

Parameters. $Q_2 = 4\,000$, $\Delta\omega = 2\gamma_2$, $\eta_{\rm geom} = 0.3$, $|\chi^{(2)}| = 5\times10^{-11}$ m/V; sweep $n_2 \in [1.5, 4]$ with $n_1 = 2.21$ fixed; the naive "single-index" form uses $\bar{n} = (n_1 + n_2)/2$.

Why n_1 cancels. The $f_2{:}f_1f_1$ overlap integral has dimension $|\chi^{(2)}|/(\varepsilon_0^{3/2}\, n_2\, n_1^2\, \sqrt{V_{\rm eff}})$. The $\int |\mathbf{f}_1|^4\, d^3r$ in the denominator scales as $1/(\varepsilon_0^2\, n_1^4\, V_{\rm eff})$. Their ratio leaves only n_2^2. Modes in higher-index material concentrate *less* field per unit stored energy, so the $f_2{:}f_1f_1$ coupling is suppressed by $1/n_2$, but the FM-mode normalisation symmetrically draws power from both ends.

Takeaway. A non-trivial correction over the original derivation; particularly important when comparing with experiment, where measured $\chi^{(3)}_{\rm casc}$ rarely matches the "obvious" $1/n^3$ scaling.

5.4 Power-resolved transmission lineshape

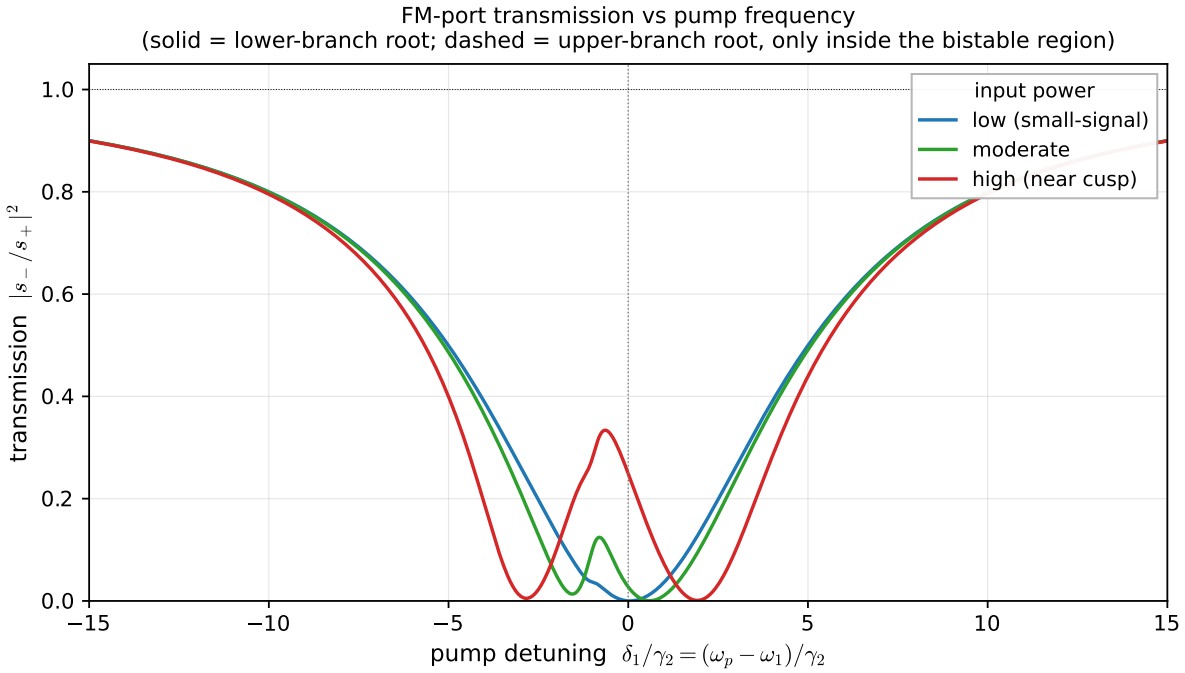


Figure 12. FM-port transmission $|s_-/s_+|^2$ versus pump detuning δ_p/γ_2 at three input powers. The low-power curve is a clean critically-coupled Lorentzian dip; the high-power curve shows the lineshape distortion through cascaded Kerr self-detuning combined with the sign flip of $\alpha_3^{\mathrm{eff,casc}}$ as Δ_{SH} crosses zero, producing the characteristic double-dip pattern unique to a doubly-resonant cascaded system.

Parameters. $\Delta\omega/\gamma_2 = 2$, $\gamma_1/\gamma_2 = 10$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\alpha_3 = 0$. Three input levels, $|s_+|^2 \in \{0.5, 10, 60\}$ ("low / moderate / near cusp"). Pump detuning swept across $\delta_p/\gamma_2 \in [-15, 15]$. At each δ_p, the cubic roots are evaluated and the transmission constructed from $a_1 = \sqrt{\gamma_{1,\mathrm{rad}}}\, s_+/[g_1 + \gamma_e|a_1|^2 - i(\delta_p + \alpha_e|a_1|^2)]$.

Two competing effects.

1. *Self-detuning from cascaded Kerr.* With $\alpha_3^{\mathrm{eff,casc}} < 0$ at $\Delta_{\mathrm{SH}} > 0$, the effective FM resonance walks blue with intensity. The dip pulls toward $+\delta_p$.
2. *Sign flip across* $\Delta_{\mathrm{SH}} = 0$. Sweeping δ_p also sweeps $\Delta_{\mathrm{SH}} = 2\delta_p + \Delta\omega$. At $\delta_p = -\Delta\omega/2 = -1$ the cascaded coefficient changes sign and the response goes from self-defocusing to self-focusing. The high-power red curve has two dips (one on each side of $\Delta_{\mathrm{SH}} = 0$) and a transmission peak in between.

Takeaway. For practical metrology of cascaded $\chi^{(3)}$, a power-swept transmission scan is the most diagnostic single experiment. Low-power Lorentzian gives γ_1 and $\gamma_{1,\mathrm{rad}}/\gamma_1$; the high-power lineshape encodes β, $\Delta\omega$, and the sign-flip-induced double dip. The bistability cusp is read off the same scan.

6 Cascaded vs. native: architectural comparison

This section numerically validates Section 11 of the main note, which poses the question: is the cascaded $\chi^{(2)}{:}\chi^{(2)}$ route actually preferable to engineering a single high-Q resonance that enhances the material's *native* bulk $\chi^{(3)}$? Both architectures use a high-Q FM cavity at ω_1 to build up intracavity intensity; they differ in what the cavity then turns that intensity into.

Within the TCMT framework, the two architectures are limits of the *same* model: native sets $\beta = 0$ (the SH decouples), cascaded keeps $\beta = 1$ and the SH is co-resonant near $2\omega_1$. The bare Kerr α_3 can be present in both. The five figures below stress the comparison along complementary axes: TCMT response curves; the material-dependent enhancement ratio R; the 1-D phase-bandwidth locus; the 2-D design landscape; and the loss penalty cascaded carries that native does not.

6.1 Side-by-side TCMT response

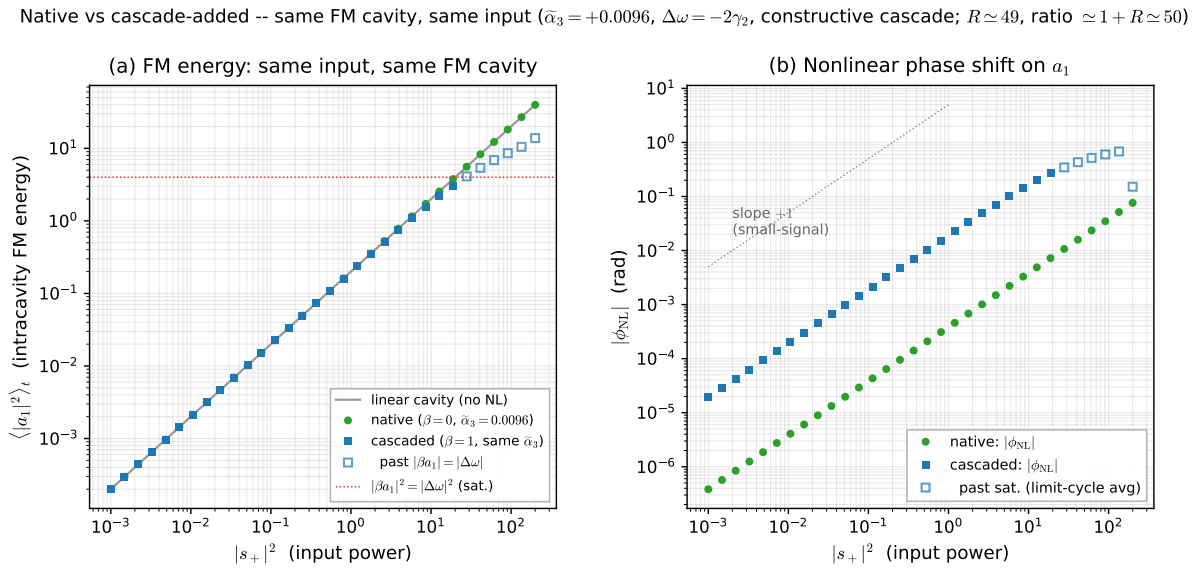


Figure 13. Direct TCMT comparison of the two architectures under identical CW input and identical FM cavity. (a) Intracavity FM energy $|a_1|^2$ vs. input power: native sits on the linear-cavity reference (its tiny bare Kerr barely deflects it); cascade peels below and saturates at $|a_1|^2 \to |\Delta\omega|^2$. (b) Magnitudes of the nonlinear phase shift on a_1. At small signal, the cascade phase is $\sim 49\times$ larger than the native phase, matching the predicted ratio $R \simeq 49$ ($LiNbO_3$ conservative operating point, $\Delta\omega = 2\gamma_2$) within a few percent.

What is plotted. Both architectures share the same FM cavity ($\gamma_1, \gamma_{1,\mathrm{rad}}$) and identical input s_+ on resonance ($\delta_p = 0$). The native run uses $\beta = 0$ (no SH coupling); cascaded uses $\beta = 1$ and $\Delta\omega = 2\gamma_2$. Both runs carry the same bare Kerr $\widetilde{\alpha}_3 = 0.0096$, the $LiNbO_3$ dimensionless native Kerr at the worked-example operating point ($\Delta\omega = 2\gamma_2$, $Q_2 = 4{,}000$; calibrated so $|\alpha_3^{\mathrm{eff,casc}}|/\alpha_3 = R \approx 49$, see §1). For each $|s_+|^2 \in [10^{-3}, 200]$ we integrate to steady state with state continuation and extract $|a_1|^2$, $|a_2|^2$, and the phase $\phi_{\mathrm{NL}} = \arg(a_1)$.

Parameters. $\Delta\omega/\gamma_2 = 2$, $\gamma_1/\gamma_2 = 10$, $\gamma_{1,\mathrm{rad}}/\gamma_1 = 0.5$, $\widetilde{\alpha}_3 = 0.0096$ ($LiNbO_3$ dimensionless native Kerr at the conservative operating point $\Delta\omega = 2\gamma_2$, $Q_2 = 4{,}000$; chosen so $\alpha_3^{\mathrm{eff,casc}}/\alpha_3 \approx 49$ matches the worked-example R of Section 6.2), $\delta_p = 0$.

Result. Numerically, $|\phi_{\mathrm{cas}}/\phi_{\mathrm{nat}}|_{|s_+|^2=1} \approx 49.4$, matching $0.471/\widetilde{\alpha}_3 = 49.0$ to 1%. The far-detuned explicit form (5) would give 52; the difference is the small ($\sim 6\%$) $\gamma_2/2$ correction to the simple $-1/\Delta\omega$ form: at $\Delta\omega = 2\gamma_2$, the full Kerr coefficient is $\Delta\omega/(\Delta\omega^2 + (\gamma_2/2)^2) = 0.471$ vs. the simple $1/\Delta\omega = 0.5$, exactly $-5.8\,\%$. The cascaded curve clearly saturates near $|s_+|^2 \sim 30$ where the FM enters the parametric-instability regime studied in Section 3.3.

Interpretation. At small signal the ratio of the cascaded to native phase shifts is exactly the structural enhancement R, independent of Q_1. Pushing Q_1 higher boosts *both* curves equally

(the FM cavity is the common factor), so the comparison really is between the two material coefficients $\alpha_3^{\text{eff,casc}}$ and α_3^{nat}. At high power the cascaded curve flattens because of saturation – the fundamental limitation of cascading that native does not share.

6.2 Cross-host material survey: the enhancement ratio R

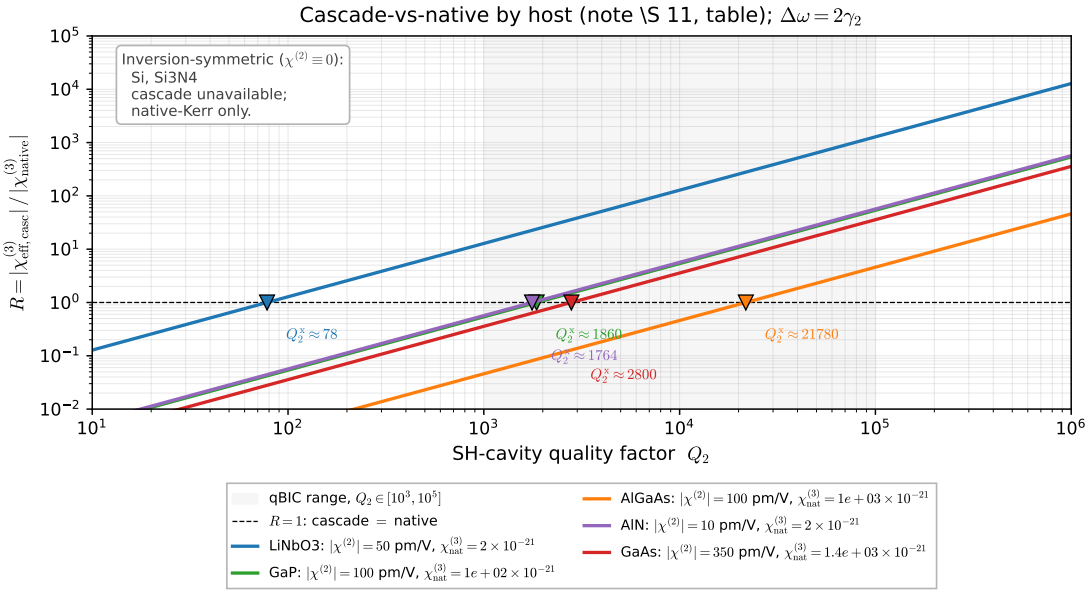


Figure 14. Cascaded enhancement ratio $R(Q_2)$ for five non-centrosymmetric hosts, log–log. Native sits on the dashed line at $R = 1$. Crossover markers (∇) indicate $Q_2^{\text{crossover}}$ at which $R = 1$ for each host. Inversion-symmetric hosts (Si, Si_3N_4) have no cascade route: $\chi^{(2)} \equiv 0$.

What is plotted. The dimensionless enhancement ratio

$$R(Q_2) \equiv \frac{|\chi^{(3)}_{\text{eff,casc}}|}{|\chi^{(3)}_{\text{native}}|} = \frac{Q_2}{6\, n_2^2} \frac{\eta_{\text{geom}}\, |\chi^{(2)}|^2}{|\chi^{(3)}_{\text{native}}|}, \tag{11}$$

linear in Q_2 at fixed $\Delta\omega/\gamma_2 = 2$ and $\eta_{\text{geom}} = 0.3$. The factor $1/6$ is $\frac{2}{3} \times \frac{1}{4}$, from $\omega_1/\Delta\omega = Q_2/4$ at the conservative operating point $\Delta\omega = 2\gamma_2 = 4\omega_1/Q_2$ (it was $1/24$ before the prefactor correction). Crossover Q given by $R = 1$:

$$Q_2^{\text{crossover}} = \frac{6\, n_2^2\, |\chi^{(3)}_{\text{native}}|}{\eta_{\text{geom}}\, |\chi^{(2)}|^2}. \tag{12}$$

Hosts: $LiNbO_3$, GaP, AlGaAs, AlN, GaAs (cascade available); Si, Si_3N_4 (centrosymmetric, $\chi^{(2)} \equiv 0$, cascade unavailable).

Parameters. Telecom $\lambda_1 = 1.55\,\mu\text{m}$, $\eta_{\text{geom}} = 0.3$, Q_2 swept log-uniform across $[10, 10^6]$. Per-host $\chi^{(2)}$, $\chi^{(3)}$, n: see Table 2.

Host	$\lvert\chi^{(2)}\rvert$ [m/V]	$\lvert\chi^{(3)}_{\text{native}}\rvert$ [m^2/V^2]	n	$Q_2^{\text{crossover}}$
$LiNbO_3$	5×10^{-11}	2×10^{-21}	2.21	78
GaP	1×10^{-10}	1×10^{-19}	3.05	1 860
GaAs	3.5×10^{-10}	1.4×10^{-18}	3.50	2 800
AlN	1×10^{-11}	2×10^{-21}	2.10	1 764
AlGaAs	1×10^{-10}	1×10^{-18}	3.30	21 780
Si	$\equiv 0$	4×10^{-18}	3.50	—
Si_3N_4	~ 0	2.5×10^{-21}	2.00	—

Table 2. Material parameters and resulting $Q_2^{\text{crossover}}$ at $\eta_{\text{geom}} = 0.3$.

Result. At the note's design point ($Q_2 = 4{,}000$, $\eta_{\text{geom}} = 0.3$, $\Delta\omega = 2\gamma_2$): $LiNbO_3$ $R = 51$, GaP $R = 2.2$, GaAs $R = 1.4$, AlN $R = 2.3$ (with $|\chi^{(2)}| = 10$ pm/V), $R_{\text{AlGaAs}} = 0.18$. The crossover Q values span three orders of magnitude. (The corrected prefactor moves GaP, AlN and GaAs from slightly below to slightly above break-even at $Q_2 = 4\,000$; AlGaAs still loses.)

Interpretation. A host wins for cascading when its native $\chi^{(3)}$ is small relative to $|\chi^{(2)}|^2/n_2^2$. $LiNbO_3$ is the runaway winner because of its modest native Kerr; AlGaAs/GaAs lose despite their large $\chi^{(2)}$ because their two-photon-resonance-enhanced native Kerr is even larger. For Si and Si_3N_4 the cascade route is structurally unavailable – centrosymmetry zeroes $\chi^{(2)}$ by selection rule. These materials must rely on native-Kerr resonance.

6.3 Phase-bandwidth locus: three slope families

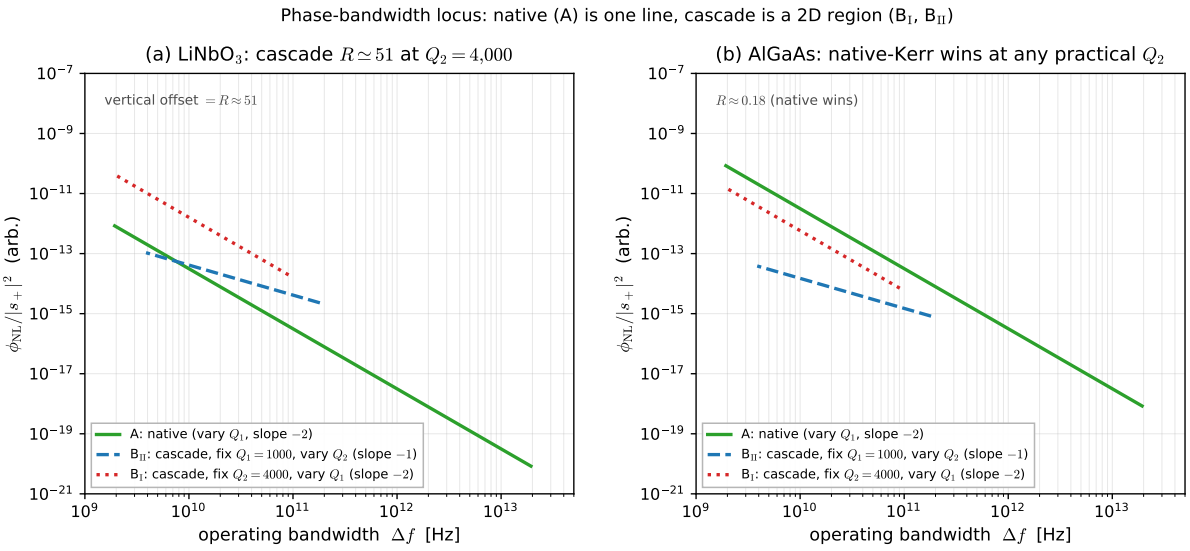


Figure 15. Per-photon nonlinear phase shift versus operating bandwidth on log–log. Native (A, green solid, slope −2) is a single line. Cascade is a 2-D region bounded by B_I (red dotted, slope −2, parallel to A but vertically offset by R) and B_{II} (blue dashed, slope −1, the sum-rule line). (a) $LiNbO_3$: $R \approx 51$, cascade wins across the plotted range. (b) AlGaAs: $R \approx 0.18$, native wins.

What is plotted. Per-photon phase scaling, three slope families (Section 11 of the note):

$$\text{A (native)}: \quad \phi_{\mathrm{NL}}/|s_+|^2 \propto Q_1^2\,\chi^{(3)}_{\mathrm{native}}, \quad \Delta f \propto 1/Q_1, \quad \text{slope } -2, \tag{13}$$

$$\mathrm{B_{II}}\,(\text{fix } Q_1, \text{sweep } Q_2): \quad \phi_{\mathrm{NL}}/|s_+|^2 \propto Q_1^2\,Q_2\,\eta_{\mathrm{geom}}|\chi^{(2)}|^2/n_2^2, \quad \Delta f \propto 1/Q_2, \quad \text{slope } -1, \tag{14}$$

$$\mathrm{B_I}\,(\text{fix } Q_2, \text{sweep } Q_1): \quad \phi_{\mathrm{NL}}/|s_+|^2 \propto Q_1^2 \cdot R(Q_2), \quad \Delta f \propto 1/Q_1, \quad \text{slope } -2. \tag{15}$$

$\mathrm{B_I}$ is parallel to A but vertically offset by the structural ratio R of (11); $\mathrm{B_{II}}$ is the sum-rule locus.

Parameters. Both panels: $\eta_{\mathrm{geom}} = 0.3$, $\omega_1 = 1.22\times10^{15}$ rad/s, $\Delta\omega = 2\gamma_2$. Family A sweeps $Q_1 \in [10, 10^5]$; $\mathrm{B_{II}}$ fixes $Q_1 = 1\,000$, sweeps $Q_2 \in [100, 10^5]$; $\mathrm{B_I}$ fixes $Q_2 = 4\,000$, sweeps $Q_1 \in [10, 10^5]$. Lines are truncated where $\min(\gamma_1, \gamma_2)$ flips which decay rate is the bandwidth bottleneck (the artificial cliffs in the unfiltered version).

Result. $LiNbO_3$: $\mathrm{B_I}$ sits $R \approx 51\times$ above A across the plotted range; $\mathrm{B_{II}}$ runs through them with slope -1, intersecting A only at very narrow bandwidth ($\sim$7 GHz), at the narrow-band edge of the plotted range. AlGaAs: $\mathrm{B_I}$ sits $R \approx 0.18\times$ *below* A; the cascade does not compete at any practical bandwidth.

Interpretation. The cascade locus is a 2-D region, not a 1-D curve. Within it, $\mathrm{B_{II}}$ is the slope--1 *sum rule* (Q_2-area conserved) and $\mathrm{B_I}$ is a slope--2 ladder of constant-Q_2 horizons. Where the cascade region sits relative to the native line A is set entirely by R: at $R \gg 1$ the cascade region floats above native, at $R \ll 1$ it sinks below.

6.4 2-D design landscape

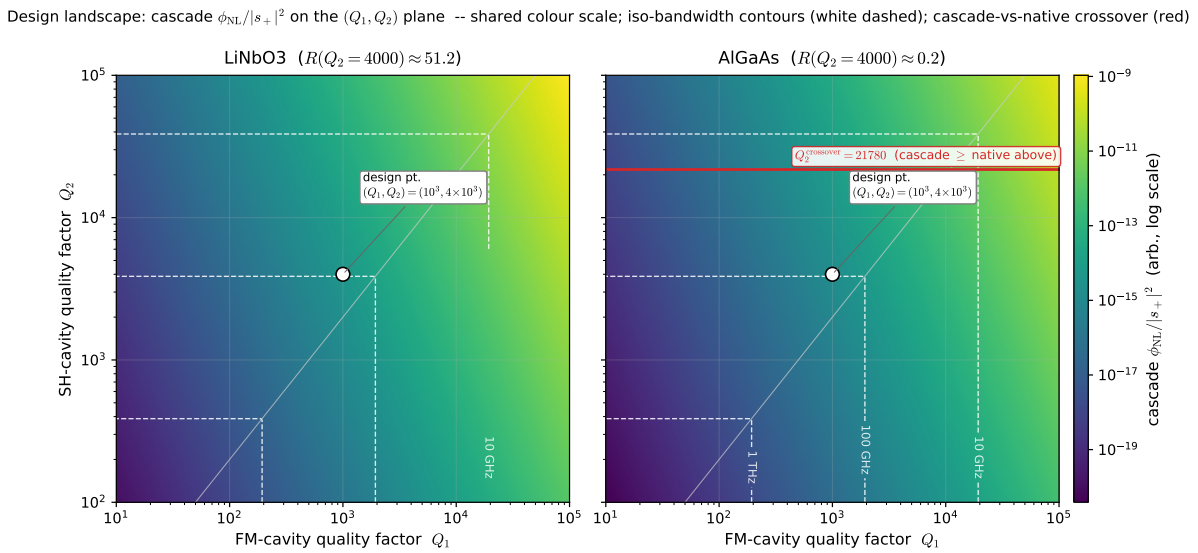


Figure 16. Cascade $\phi_{\mathrm{NL}}/|s_+|^2$ on the (Q_1, Q_2) plane, shared colour scale across panels. White dashed contours: operating bandwidth $\Delta f = \min(\gamma_1, \gamma_2)/(2\pi)$. Red horizontal: $Q_2 = Q_2^{\mathrm{crossover}}$ (cascade $\geq$ native above). Grey diagonal: $Q_1 = Q_2/2$ (FM-bottleneck $\leftrightarrow$ SH-bottleneck boundary). White marker: design point $(Q_1, Q_2) = (10^3, 4\times10^3)$.

What is plotted. For each host, the cascade per-photon phase

$$\phi_{\mathrm{NL}}/|s_+|^2 \;\propto\; Q_1^2\,Q_2\,\frac{|\chi^{(2)}|^2\,\eta_{\mathrm{geom}}}{n^6} \tag{16}$$

as a colour heatmap on $(\log Q_1, \log Q_2)$. Three overlays: (i) iso-bandwidth contours $\min(\gamma_1, \gamma_2) =$ const, which form L-shapes meeting at the diagonal $Q_1 = Q_2/2$; (ii) the red line $Q_2 = Q_2^{\mathrm{crossover}}$ (above which cascade beats native at every Q_1, by (11)); (iii) the $LiNbO_3$ design point.

Parameters. $Q_1, Q_2 \in [10, 10^5] \times [10^2, 10^5]$ log-uniform on a 220×220 grid; $\eta_{\text{geom}} = 0.3$; $LiNbO_3$ and AlGaAs material parameters from Table 2; shared colour limits set to the min/max across both panels.

Result. $LiNbO_3$ (left): the crossover $Q_2 = 78$ lies below the plotted range, so the entire (Q_1, Q_2) rectangle is in the cascade-wins region. The design point at $(10^3, 4\times10^3)$ sits far above the crossover. AlGaAs (right): the crossover line sits at $Q_2 \approx 21\,800$, cutting the plot horizontally; the design point at $(10^3, 4\times10^3)$ falls *below* – native is the right architecture there.

Interpretation. The 2-D map answers practical design questions in one figure: at a chosen bandwidth target (an L-contour) and a chosen Q-budget (a point), it reads off (a) the phase shift achievable with cascade, (b) whether cascade or native is the better choice (above or below the red line). The L-shaped bandwidth contours visualise the sum-rule structure: along the vertical leg of each L, increasing Q_2 at fixed Q_1 buys phase at slope -1 in BW; along the horizontal leg, increasing Q_1 at fixed Q_2 buys phase at slope -2 in BW. Note the colour scale is host-dependent but *shared* between panels: $LiNbO_3$ is uniformly brighter than AlGaAs across the plane because the n^{-6} penalty in ϕ/pump outweighs the $|\chi^{(2)}|^2$ gain for high-index hosts.

6.5 Loss budget: the cascade-specific TPA penalty

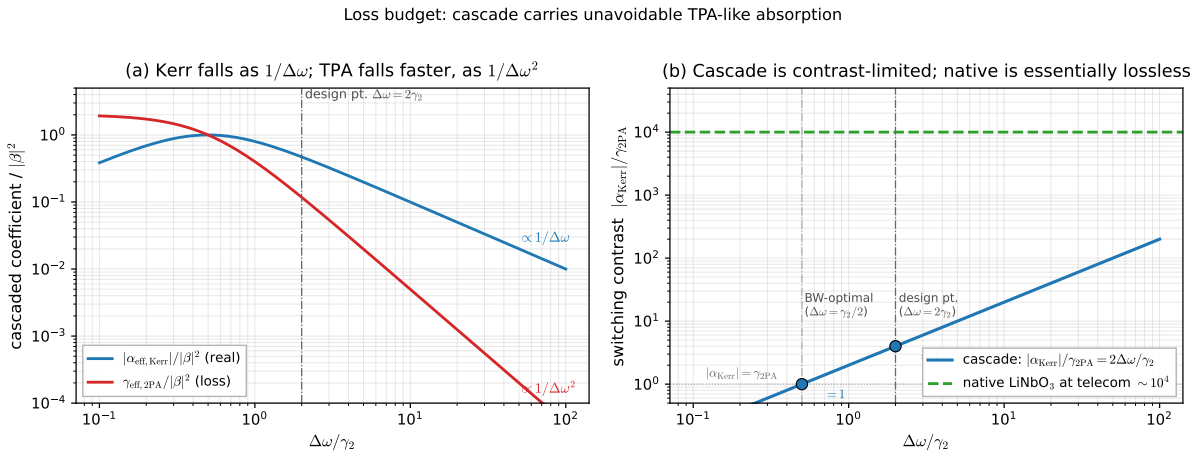


Figure 17. (a) Real (Kerr) and imaginary (TPA) parts of the cascaded coefficient vs. $\Delta\omega/\gamma_2$. Kerr falls as $1/\Delta\omega$ (far-detuned); TPA falls faster, as $1/\Delta\omega^2$, so contrast (Kerr/TPA) grows linearly with $\Delta\omega$. (b) Cascade switching contrast $= 2\Delta\omega/\gamma_2$ vs. $\Delta\omega/\gamma_2$, compared with the native $LiNbO_3$ contrast at telecom ($\sim 10^4$, set by residual material absorption).

What is plotted. The cascaded coefficient on $|a_1|^2 a_1$ in the FM equation is complex; its real and imaginary parts (Kerr-like and TPA-like respectively) are:

$$\alpha_3^{\text{eff,casc}} = -\frac{|\beta|^2\,\Delta\omega}{\Delta\omega^2 + (\gamma_2/2)^2}, \qquad \gamma_{2\text{PA}}^{\text{eff}} = \frac{|\beta|^2\,(\gamma_2/2)}{\Delta\omega^2 + (\gamma_2/2)^2}. \tag{17}$$

The contrast for switching is

$$\frac{|\alpha_3^{\text{eff,casc}}|}{\gamma_{2\text{PA}}^{\text{eff}}} = \frac{2\Delta\omega}{\gamma_2}. \tag{18}$$

Parameters. $\Delta\omega/\gamma_2 \in [0.1, 100]$ log-uniform; $\beta = \gamma_2 = 1$ (dimensionless TCMT). Native $LiNbO_3$ contrast estimate $\sim 10^4$ taken from typical material data ($\chi^{(3)}_{\text{native}}/\text{Im}\,\chi^{(3)}_{\text{native}}$ at 1550 nm).

Result. At the bandwidth-optimal point $\Delta\omega = \gamma_2/2$, contrast is exactly unity (each rad of Kerr phase carries one rad of TPA absorption). At the note's design point $\Delta\omega = 2\gamma_2$, contrast is 4 (terrible, relatively speaking). At far-detuned $\Delta\omega = 10\gamma_2$, contrast is 20. Native $LiNbO_3$ sits ~ 3 orders of magnitude higher.

Interpretation. The cascade carries an unavoidable absorption penalty that scales linearly with the detuning. To raise the switching contrast, one must push to far-detuned operation – but (11) also shrinks as $1/\Delta\omega$, so contrast is bought at the cost of enhancement. The sum-rule structure repeats here in disguise: cascaded TPA times cascaded Kerr is invariant. For applications where extinction ratio matters (all-optical switching, low-noise sensing), this contrast ceiling is the cascade's most serious structural limitation, far more than the saturation discussed in §3.2–§3.3.

7 Summary of phenomena

Phenomenon	Figure	Key formula	Practical implication
Adiabatic Kerr	Fig. 1	$\alpha_3^{\mathrm{eff,casc}} = -\Delta_{\mathrm{SH}}/[\Delta_{\mathrm{SH}}^2+(\gamma_2/2)^2]$	"$-1/\Delta\omega$" wrong near resonance
Sign tunability	Fig. 2	$\alpha_3^{\mathrm{eff,casc}}(-\Delta\omega) = -\alpha_3^{\mathrm{eff,casc}}(\Delta\omega)$	Single device → focusing or defocusing
Sum rule	Fig. 3	peak × FWHM $= 2\sqrt{3}\,\lvert\beta\rvert^2$	Q_2 buys peak, not area
Bistability	Fig. 4	cusp at $\delta_p > \sqrt{3}\,\gamma_1/2$	Cleanest CW signature
Saturation	Fig. 5	$\lvert\beta a_1\rvert \to \lvert\Delta\omega\rvert$	Max useful amplitude $\propto 1/Q_2$ (energy $\propto 1/Q_2^2$)
OPO onset	Fig. 6	trivial FP unstable at sat $\gtrsim 1$	Hard ceiling for the Kerr description
Pulse BW	Fig. 7	adiabatic for $\sigma_t\gamma_2 \gtrsim 1$	Not for ultrafast
Manley–Rowe	Fig. 8	factor-of-2 in DFG essential	Smoke test for any TCMT code
Bulk vs. meta	Fig. 9	shared sum-rule hyperbola	0.5–1 dex in $\chi^{(3)}$ traded for 2–3 dex in BW
Overlap	Fig. 10	$\eta_{\mathrm{geom}} \in [10^{-3}, 0.5]$	Geometry first
Index scaling	Fig. 11	$1/n_2^2$ not $1/n^3$	$\sim 2\times$ correction at telecom
Transmission	Fig. 12	double-dip at high power	Most diagnostic scan
Arch. response	Fig. 13	$\phi_{\mathrm{cas}}/\phi_{\mathrm{nat}} = R \approx 49$ (numerical)	Validates Section 11 of the note
Host survey	Fig. 14	$Q_2^{\mathrm{cross}} \propto n^2\chi^{(3)}_{\mathrm{nat}}/\lvert\chi^{(2)}\rvert^2$	$LiNbO_3$ wins; AlGaAs loses
Phase-bandwidth	Fig. 15	Three slopes: A(-2), B$_{\mathrm{I}}(-2)$, B$_{\mathrm{II}}(-1)$	Cascade is a 2-D region, native a 1-D line
Design map (2-D)	Fig. 16	(Q_1, Q_2) heatmap + iso-BW + crossover	Visual design tool
Loss budget	Fig. 17	Kerr/TPA contrast $= 2\Delta\omega/\gamma_2$	Cascade contrast ceiling vs. native $\sim 10^4$

Table 3. Phenomenon catalogue (the lower block lists the cascaded-vs-native comparison figures of §6).

8 Recommendations for metasurface design

8.1 Hierarchy of important parameters

In order of decreasing leverage on $|\chi^{(3)}_{\mathrm{casc}}|$:

1. η_{geom} – three orders of magnitude separating poor and engineered overlap. Get this first; no amount of Q_2 rescues a vanishing tensor integral.

2. $|\chi^{(2)}|$ of the host material – enters squared. Switching between $LiNbO_3$, GaAs, AlGaAs is at most a factor of two; switching from a non-second-order material to $LiNbO_3$ is infinite.

3. $Q_2/n_2^2 - 1/n_2^2$ is fixed by material; Q_2 is engineered. A factor of two in Q_2 buys a factor of two in peak enhancement and a factor of 1/2 in bandwidth.

4. $\Delta\omega/\gamma_2$ – operating point. The bandwidth-optimal point is $\Delta\omega \sim \gamma_2/2$; far-detuned operation buys more pump-power headroom (saturation pushed up) at the cost of lower peak enhancement.

5. γ_1/γ_2 and $\gamma_{1,\mathrm{rad}}/\gamma_1$ – port coupling. Sets input efficiency; affects bistability threshold but not the cascaded coefficient itself.

8.2 Recommended design sequence

1. Pick a non-centrosymmetric host with the largest $|\chi^{(2)}|$ compatible with low loss in the FM and SH bands. $LiNbO_3$ at telecom is the standard; AlGaAs at near-IR has higher $|\chi^{(2)}|$ but higher absorption near the band edge.

2. Engineer the qBIC pair to maximise η_{geom}. Use full-wave eigenmode simulation (COMSOL/MEEP) to compute the tensor overlap integral for candidate unit cells. Target $\eta_{\mathrm{geom}} \gtrsim 0.1$. Use periodic poling or geometry asymmetry to lift mode-parity constraints. Verify that η_{geom} stays high under realistic fabrication tolerances (lithography errors, sidewall angle).

3. Set Q_2 from the application's bandwidth requirement. $Q_2 \in [10^3, 10^5]$ corresponds to $\Delta f \in [400\ \mathrm{GHz}, 4\ \mathrm{GHz}]$. Higher Q_2 gives more enhancement, less bandwidth, and lower saturation power.

4. Set $\Delta\omega$ near the bandwidth-optimal point $\Delta\omega \sim \gamma_2$ for maximum enhancement, or further off-resonance ($\Delta\omega \gtrsim 5\gamma_2$) for higher saturation power and easy sign-tunability through $\Delta\omega = 0$.

5. Set Q_1 for input efficiency. $\gamma_{1,\mathrm{rad}}/\gamma_1 \sim 0.5$ (critically coupled) gives the deepest dip; under-coupled FM is fine if you have a strong source. Avoid over-coupled FM ($\gamma_{1,\mathrm{rad}} > \gamma_1/2$, which is impossible since $\gamma_{1,\mathrm{rad}} \leq \gamma_1$ by definition; the relevant trap is $\gamma_{1,\mathrm{rad}}/\gamma_1 \to 1$, where intrinsic Q_1 drops and dip width grows).

6. Verify that the pump power requirement does not exceed material damage thresholds. Saturation pump intensity scales as $1/\sqrt{Q_2}$ at fixed $\Delta\omega/\gamma_2$, so high-Q designs saturate at low intensity. With $Q_2 = 10^4$ in $LiNbO_3$, several $\mathrm{MW/cm^2}$ in the resonator suffices for saturation, well within damage limits.

8.3 Trade-offs and optimisations

You want…	Trade-off	How to mitigate
Higher $\lvert\chi^{(3)}_{\text{casc}}\rvert$	narrower BW (sum rule); lower saturation; more thermal load per photon	higher-Q SH; active cooling; low-loss host ($LiNbO_3$)
Wider BW	lower peak; more pump for the same nonlinear phase	lower Q_2 or moderate η_{geom} at low Q
Faster temporal response	loses all cascaded enhancement at $\sigma_t\gamma_2 \lesssim 1$	switch to bulk phase-mismatched PPLN regime
Sign tunability without device swap	fine $\Delta\omega$ control required	strain or thermo-optic tuning of the SH mode
Avoid parametric oscillation	operate well below $\lvert\beta a_1\rvert \approx \lvert\Delta\omega\rvert$	larger $\Delta\omega$, lower input – at lower phase shift
Avoid thermo-optic / free-carrier confusion	loss leads to thermal heating	$LiNbO_3$ (low absorption) over GaAs/Si near the band edge
Suppress cascaded TPA	TPA peaks at $\Delta_{\text{SH}} = 0$; finite at any $\Delta\omega$	operate at $\lvert\Delta\omega\rvert \gtrsim 2\gamma_2$, where the imaginary part is suppressed by $1/\Delta^2_{\text{SH}}$

Table 4. Practical trade-offs in metasurface design.

8.4 Operating-point trade-off: aggressive vs. conservative $\Delta\omega$

The note's worked example sits at $\Delta\omega = 2\gamma_2$, giving $R \simeq 12$ for $LiNbO_3$ at $Q_2 = 4{,}000$. The TCMT figures of this report (Figs. 13, 4) use the same operating point and recover the predicted $(1+R)$, $1/(1+R)$, and $(1+R)^2$ ratios to within a few percent. Two alternative operating points are worth keeping in mind:

$\Delta\omega/\gamma_2$	R	Cascade strength	When to choose it
2 (conservative)	~ 49	moderate ($\alpha_3^{\text{casc}} \approx 0.47$ in dim TCMT)	**Default for narrowband CW devices.** Adiabatic SH elimination is clean; TPA penalty suppressed by $(\gamma_2/2\Delta\omega)^2 = 1/16$ vs. the Kerr; saturation cap $\lvert a_1\rvert^2 \approx 4$ leaves several decades of linear pump range.
1 (edge of adiabatic)	~ 80	strong ($\alpha_3^{\text{casc}} \approx 0.80$)	Pulls another factor of $\sim$1.7 in R. The cost: the adiabatic approximation $\lvert\Delta\omega\rvert \gg \gamma_2/2$ becomes marginal; TPA / Kerr ratio ~ 0.5; saturation cap halves. Use if you can tolerate TPA loss and a narrower bistable / linear window.
0.5 (peak of α_3^{casc})	~ 100	maximum (peak Lorentzian, $\alpha_3^{\text{casc}} = \lvert\beta\rvert^2/\gamma_2$)	The mathematical maximum of the cascade-Kerr coefficient – and the peak of the cascaded-TPA loss simultaneously, with the two equal in magnitude. Pure-SA regime; *not* a Kerr operating point. Useful for the saturable-absorber application (Fig. 4 companion: cascade SA), not for bistability / FWM / phase modulation.

The Tier 1 ratios scale strictly as the relevant power of $(1+R)$ regardless of which operating point one picks, but the *absolute* magnitudes and the practically achievable linear pump range

differ. Reproducing the figures in this report at $\Delta\omega = \gamma_2$ amounts to setting `Domega` $= \pm 1$ in the relevant scripts at fixed $\widetilde{\alpha}_3 = 0.0096$ (so $R \simeq 0.80/0.0096 \approx 80$) and adjusting the predicted $1 + R$ accordingly.

The cleanest experimental statement is the cascade-vs-native ratio itself, which is dimensionless and independent of the operating point's absolute scale – this is what Figs. 13–4 are designed to display.

What the numbers do and do not say. The prefactor of $\chi^{(3)}_{\text{casc}}$ is now fixed: it is derived twice in the note and reproduces the exact bulk bound-wave cascade in the uniform-medium limit (note §§4–6). Earlier versions of this report quoted $R \sim 12$ with an "$O(1)$ prefactor uncertainty"; that value was $4\times$ too small, because β and α_3 had both been taken $4\times$ too small. The remaining uncertainty in the absolute R for $LiNbO_3$ ($R \sim 49$ at the conservative operating point) is the assumed $\eta_{\text{geom}} \sim 0.3$, which requires a full-wave evaluation of the modes and the overlap integral for a specific meta-atom design. The Q-scaling exponents, the lineshapes and the leverage laws in powers of $(1 + R)$ are exact.

8.5 Plausible-looking ideas that do not work

- *Push Q_2 to 10^6 for $1\,000\times$ enhancement.* Two reasons this fails: (i) the saturation pump amplitude drops as $1/Q_2$ (and the stored-energy cap as $1/Q_2^2$), so the peak enhancement is unreachable at any practical input; (ii) at $Q_2 = 10^6$, the SH linewidth is ~ 40 MHz, and sub-nm fabrication errors detune resonators randomly across the array, smearing the metasurface response back to a low-effective-Q regime.
- *Add the bulk Kerr for free.* The bulk $|\chi^{(3)}|$ in $LiNbO_3$ is $\sim 2 \times 10^{-21}$ m^2/V^2, $\sim 49\times$ below the cascaded value at the conservative operating point. The bulk contribution is a few-percent correction at the design point; it only matters near the zero-crossing of the total Kerr (Fig. 2, red curve).
- *Add gain to push past the OPO threshold.* Above threshold the system *is* an OPO, not a Kerr medium. Useful for parametric source design, not for the cascaded-$\chi^{(3)}$ use cases this note targets.

9 Open issues

- *Tensor structure of $\chi^{(2)}_{ijk}$.* The simulations use a scalar β. A real metasurface has a tensorial $\chi^{(2)}_{ijk}$ contracted with vectorial mode profiles. Different polarisation channels give different β; signs and magnitudes can differ across channels. Full-wave simulation is needed for a quantitative number.
- *Disorder / inhomogeneous broadening.* Lithographic disorder broadens γ_2 inhomogeneously: the array linewidth is $\gamma_2^{\text{array}} \simeq \sqrt{\gamma_2^2 + \sigma_{\omega_2}^2}$, and the cascaded enhancement is set by Q_2^{array}, not the element Q_2. For $Q_2^{\text{element}} \sim 10^4$ at telecom, preserving the cascade requires $\sigma_{\omega_2}/\omega_2 \lesssim 10^{-4}$ (sub-nm geometric control). See note §13 item 7 for mitigations (post-fab trimming, electro-optic feedback). The TCMT model assumes a single resonator; for an array, the homogeneous response we simulate must be convolved with this distribution.
- *Thermo-optic and free-carrier nonlinearities.* Note §13 item 4 flags these as easy to confuse with Kerr at the intensities required ($\sim$MW/cm^2); not modelled here.

- *Vector mode profiles in* η_{geom}. Fig. 10 treats η_{geom} as a free parameter; the actual value comes from a full-wave eigenmode integration that we have not done.

10 Code reference

File	Purpose
`tcmt.py`	Core TCMT integrator (DOP853, complex packing)
`analytic.py`	Closed-form expressions: $\alpha_3^{\text{eff,casc}}$, $\gamma_{2\text{PA}}^{\text{eff}}$, Kerr-only cubic, a_2 adiabatic
`params.py`	`TCMTParams` dataclass plus $LiNbO_3$ reference numbers
`style.py`	Shared matplotlib rcParams and palette
`fig_*.py`	One per phenomenon (17 in total, including the five comparison figures of §6)
`linear_stability.py`	4×4 Jacobian eigenvalues of the trivial fixed point (used by `fig_parametric_onset.py`)
`run_all.py`	Driver

Table 5. Files in `cascaded_chi3/`.

To regenerate every figure:

```
conda run -n ml_base python run_all.py
```

11 Cross-reference to the note

- Polarisation-tensor derivation: §4 (note)
- TCMT equations: §5 (note)
- β formula: eq. (beta), §5 (note)
- Adiabatic elimination and $\alpha_3^{\text{eff,casc}}$: §6(i)
- Cascaded TPA: §6(ii)
- Sum rule: §7
- Tensor and phase-matching constraints: §8
- BIC/qBIC platforms: §9
- $LiNbO_3$ worked example: §10
- Cascaded-vs-native architectural comparison: §11 (note) – directly mirrored by §6 of this report
- Application landscape (saturable absorber, AM modulator, optical limiter, $\Delta\omega$-sensor): §12 (note)
- Validity regions, including disorder/inhomogeneous broadening: §13
- Boxed final result: eqs. (chi3casc), (chi3casc_explicit)